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PulseLM: A Foundation Dataset and Benchmark for PPG-Text Learning

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Abstract

Photoplethysmography (PPG) is a widely used non-invasive sensing modality for continuous cardiovascular and physiological monitoring across clinical, laboratory, and wearable settings. While existing PPG datasets support a broad range of downstream tasks, they typically provide supervision in the form of numerical measurements or task-specific labels, limiting their suitability for language-based physiological reasoning and multimodal foundation models. In this work, we introduce PulseLM, a large-scale PPG-text dataset designed to bridge raw PPG waveforms and natural language through a unified, closed-ended question answering (QA) formulation. PulseLM aggregates PPG recordings from fifteen publicly available sources and harmonizes heterogeneous annotations into twelve common physiologically QA tasks. The dataset comprises 1.31 million standardized 10-second PPG segments, associated with 3.15 million question-answer pairs. We further define reproducible preprocessing, supervision, and evaluation protocols and establish baseline benchmarks using multimodal PPG-aware large language models. PulseLM provides a standardized foundation for studying multimodal physiological reasoning, cross-dataset generalization, and scalable benchmarking of PPG-based language models. The data and code can be found publicly available at: https://github.com/manhph2211/PulseLM.

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CCS Concepts

• **Applied computing** → **Health informatics**.

Keywords

PPG-Text Dataset, PPG Question Answering, PPG Language Models

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1 Introduction

Photoplethysmography (PPG) is a non-invasive optical sensing modality that underpins a wide range of modern health monitoring systems, from clinical pulse oximeters [18] to consumer wearables such as smartwatches and earbuds [1–4]. In principle, PPG measures changes in blood volume within the peripheral vasculature by emitting light onto the skin using a light-emitting diode (LED) and detecting variations in the intensity of transmitted or reflected light with a photodetector, where cardiac-induced pulsatile blood flow gives rise to a characteristic waveform. From that, PPG waveforms can be analyzed to enable continuous estimation of a broad range of cardiovascular and respiratory indicators, including heart rate (HR) [34], heart rate variability (HRV) [13, 42], respiratory rate (RR) [21], blood pressure (BP) [19], stress levels [38], and sleep-disordered breathing (SDB) [10]. Owing to its low cost, ease of deployment, and applications, PPG has become one of the most widely collected physiological signals worldwide.

From a machine learning perspective, this abundance of PPG data presents a significant opportunity for developing physiological models that can operate across tasks, devices, and recording environments. In fact, recent advances in deep learning models have demonstrated their effectiveness on diverse PPG downstream tasks [5, 30, 32, 37]. That said, the predominant use of machine

learning for PPG signals remains constrained by task-specific supervision. Specifically, most existing PPG datasets provide numerical measurements or categorical labels tied to a single downstream task, such as heart rate regression [34] or blood pressure classification [15]. This supervision paradigm encourages models to learn narrow input-output mappings rather than higher-level, semantically grounded representations of physiological state. As a result, such models often struggle to generalize across tasks and offer limited interpretability or interaction beyond fixed prediction outputs.

This limitation becomes increasingly pronounced in the context of modern foundation models. In domains such as natural language processing, computer vision, and medical imaging, large language models (LLMs) and multimodal models have demonstrated that language can serve as a unifying interface for reasoning, explanation, and cross-task generalization [22, 33, 39]. In physiological sensing, similar advances have been enabled by large-scale electrocardiogram (ECG)-text and image-report datasets, which allow models to align raw signals with clinical semantics expressed in natural language [27, 31, 36, 44]. However, comparable resources for raw PPG signals are notably absent, preventing the development and evaluation of models that can interpret, explain, or reason over PPG signals using natural language.

One promising way to bridge this gap is to formulate PPG understanding as an LLM-based question-answering (QA) problem. Question answering provides a natural and flexible interface that mirrors how clinicians and end users interact with physiological data, for example, by asking whether a heart rate is normal, whether signal quality is sufficient, or whether a recording indicates elevated cardiovascular risk. Importantly, a QA formulation can unify heterogeneous downstream tasks under a single supervision paradigm, avoiding the need for task-specific regression heads while enabling models to associate continuous waveform patterns with discrete, interpretable physiological concepts.

However, enabling PPG-centric question answering at scale poses substantial challenges. Specifically, *public PPG datasets are highly fragmented, differing in sensor placement (fingertip vs. wrist vs. ear), sampling rate (60 Hz vs. 128 Hz), recording environment (clinical vs. in-the-wild, stationary vs. motion) , subject population (patients vs healthy, young vs. elderly people), and available annotations (heart rate, stress level, motion status, etc.). Moreover, none of these datasets natively provide natural-language descriptions or QA supervision.* As a result, there is currently no large-scale, standardized benchmark that supports language-based reasoning over PPG signals across diverse physiological domains and real-world conditions.

To address this gap, we introduce **PulseLM**, a large-scale PPG-Text QA dataset and benchmark designed to support multimodal PPG learning and interpretation. PulseLM aggregates and harmonizes PPG recordings from diverse public sources spanning clinical, laboratory, and in-the-wild environments. For each standardized PPG segment, we derive physiologically grounded categorical labels from existing annotations or co-recorded reference signals (e.g., ECG) and convert them into closed-ended QA supervision using controlled natural-language question pools. The resulting dataset comprises over one million PPG-Text samples and three million question-answer pairs covering a broad range of cardiovascular, respiratory, autonomic, behavioral, and signal quality tasks.

In addition to releasing the dataset, we establish a unified benchmarking protocol for PPG question answering and provide baseline results using multimodal models that integrate PPG encoders with instruction-tuned language models. By framing PPG interpretation as a language-based QA problem, PulseLM enables systematic evaluation of physiological reasoning across tasks, datasets, and recording conditions, and lays the foundation for future research on language-enabled foundation models for PPG signals.

This work makes the following contributions:

- To the best of our knowledge, we introduce the first large-scale PPG-Text QA dataset, spanning clinical, laboratory, and in-the-wild settings that comprises over three million closed-ended question-answer pairs.
- We define a corresponding benchmarking performance in multimodal PPG question answering tasks with twelve question types using consolidated PPG-language models.
- We will release a unified data processing pipeline, model training, and evaluation protocols to facilitate future research on language-based PPG understanding.

2 Background and Motivation

2.1 PPG Downstream Tasks

PPG measures volumetric changes in blood within peripheral tissue by illuminating the skin with a light source and recording variations in the reflected or transmitted light. Owing to this optical measurement principle, the resulting waveform consists of a slowly varying DC component, reflecting static tissue absorption and average blood volume, and a superimposed AC component, corresponding to pulsatile blood flow induced by cardiac activity. Under stable clinical conditions, such as fingertip recordings with minimal motion, the AC component exhibits quasi-periodic cardiac cycles with well-defined morphological features, including a dominant systolic peak and a secondary diastolic peak associated with vascular reflections. However, in practical wearable scenarios, factors such as user movement [41] or loose contact [12] between a wrist-worn device and the skin can disrupt sensor–skin coupling and deform underlying tissue, leading to attenuation or corruption of the AC component, amplification of DC drift, severe waveform distortion, or even the disappearance of characteristic pulsatile peaks.

However, with the development of signal processing techniques [9, 29] and, more recently, deep learning methods [17, 32, 34, 43], PPG has enabled a broad spectrum of downstream tasks, including heart rate (HR), heart rate variability (HRV), and respiratory rate (RR) estimation, which are now routinely deployed in consumer wearables. These tasks rely primarily on temporal characteristics of the waveform and form the foundation of many real-world PPG-based systems. Beyond those core metrics, more complex inference tasks depend heavily on subtle waveform morphology. Blood pressure (BP) estimation[11, 28], including both systolic and diastolic pressure, has been extensively studied using PPG-derived features/representation, either through regression to continuous values or classification into clinically meaningful categories (e.g., hypertension stage 1).

In a related vein, rhythm-level cardiac inference, such as atrial fibrillation (AF) and arrhythmia detection, exploits irregular inter-beat

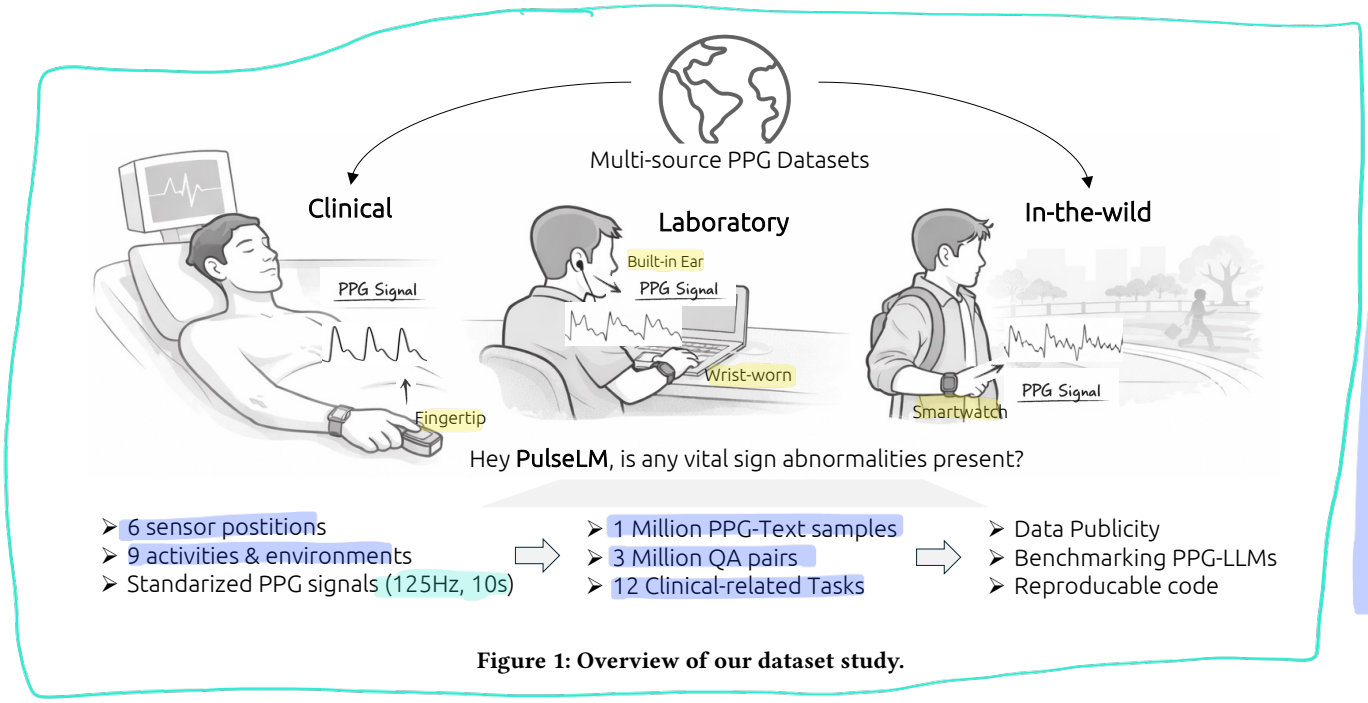

Figure 1: Overview of our dataset study.

intervals, variable pulse amplitudes, and disrupted waveform morphology captured by PPG. These tasks require models to capture fine-grained morphological and rhythm structure and longer-range temporal dependencies within the signal. More recently, PPG has been explored in a range of non-traditional and higher-level inference domains. Studies have demonstrated its utility for emotion (stress) assessment[38], sleep-disordered breathing (SDB)[10] detection. Unlike classical cardiovascular metrics, these tasks often require integrating physiological signals with contextual reasoning, temporal abstraction, and domain knowledge. As a result, successful inference in these domains extends beyond purely signal-level pattern recognition and demands more expressive representations of physiological states.

2.2 Existing PPG Datasets

To date, a wide range of publicly available PPG datasets have been released over the past decades to support research in health monitoring and wearable sensing, spanning clinical, laboratory, and in-the-wild recording environments. For example, clinical datasets are the most common category, collected in hospitals or clinics daily for a large number of patients, and typically use fingertip pulse oximeters to acquire high-fidelity PPG signals with minimal motion interference. Laboratory datasets extend this setting by adopting wearable form factors, most commonly wrist-worn or built-in ear devices[26, 38], and introducing structured protocols or controlled activities (e.g., mild physical movement such as sitting or talking) to study physiological responses under reproducible and specialized conditions. In contrast, in-the-wild or daily activity datasets[25] further improve ecological validity by capturing longer-term PPG recordings during unconstrained everyday activities (e.g., running, playing sports) using consumer-grade wearables, at the cost of increased motion artifacts and sensing variability.

Despite the diversity of available PPG datasets and their applications, existing resources exhibit several fundamental limitations when viewed through the lens of scalable physiological reasoning. Most datasets are designed around a limited downstream task and provide supervision in the form of numerical values or categorical labels specific to that task, such as heart rate labels (e.g., 65 bpm) for heart rate estimation tasks. Consequently, models trained on these datasets are typically optimized for narrow objectives and lack a unified interface for cross-task inference.

A second limitation arises from dataset fragmentation. Public PPG datasets differ substantially in sensor placement, sampling rate, signal length, preprocessing pipelines, and label definitions. Even for ostensibly similar tasks, such as heart rate variability estimation, different datasets rely on distinct metrics, time windows, or reference standards. This heterogeneity makes it difficult to combine datasets or evaluate models under consistent protocols, and often leads to models that implicitly overfit to dataset-specific characteristics rather than learning generalizable physiological representations.

Finally, and most critically for language-enabled modeling, existing PPG datasets lack natural-language supervision. Unlike ECG or medical imaging domains, where large-scale signal-text or image-report datasets [36] have enabled models to learn clinically meaningful semantics through language, PPG datasets remain confined to numeric or concretized categorical labels. This absence of language-based supervision precludes the development of models that can describe waveform characteristics, explain physiological states, or interact with users through natural-language queries.

Taken together, these limitations suggest that progress in PPG-based learning is constrained not only by modeling approaches but also by the structure and scope of available datasets. Addressing these challenges requires a reformulation of how PPG supervision is represented and consumed by learning systems.

2.3 QA as a Unifying PPG Inference

In this context, question answering (QA) offers a principled framework for bridging raw physiological signals and high-level semantic interpretation. Rather than training separate models for each downstream task, a QA formulation allows heterogeneous physiological attributes to be expressed as questions with interpretable answer choices. This mirrors how clinicians and end users naturally reason about physiological data, for example by asking whether a heart rate is within a normal range, whether a signal is sufficiently clean for interpretation, or whether a recording indicates health risk.

From a modeling perspective, QA provides several advantages: 1) it enables a unified supervision interface across diverse tasks, eliminating the need for task-specific output heads and loss functions; 2) when answer spaces are constrained to closed-ended categorical choices, QA reduces ambiguity and supports objective evaluation using standard classification metrics; 3) By associating continuous waveform patterns with discrete semantic concepts, QA also encourages models to learn latent physiological representations that are both interpretable and transferable across tasks.

Crucially, a QA-based formulation aligns naturally with multimodal large language models. By conditioning language generation or selection on raw PPG inputs, such models can learn to ground linguistic responses in physiological signals, enabling richer forms of interaction and reasoning. However, realizing this potential requires datasets that provide large-scale, physiologically grounded QA supervision across diverse recording conditions.

Motivated by these observations, we design PulseLM to reformulate PPG understanding as a closed-ended question-answering problem. By harmonizing heterogeneous public datasets and converting existing physiological annotations into structured QA supervision, PulseLM aims to provide the missing foundation for language-enabled PPG modeling. In the following section, we describe the construction of this dataset in detail, including data sources, ground truth harmonization, and QA generation procedures.

3 PulseLM Dataset

3.1 Overview

We introduce PulseLM, a large-scale PPG-Text question-answering dataset designed to enable language-based physiological reasoning across diverse tasks and recording conditions (see Figure 1). The dataset aggregates PPG recordings from 15 publicly available sources and reformulates their existing annotations into a unified, closed-ended QA representation. In total, PulseLM contains 1,310,665 standardized PPG segments paired with 3,153,899 physiologically grounded question-answer pairs spanning thirteen unique QA categories. Each PPG segment is represented as a fixed-length 10-second window and may be associated with multiple QA pairs depending on the availability of annotations in the source dataset. By expressing heterogeneous downstream tasks under a shared QA interface, PulseLM enables systematic benchmarking of multimodal models on physiological reasoning rather than task-specific prediction. To this end, the dataset spans clinical monitoring, controlled laboratory studies, and in-the-wild wearable recordings, with PPG signals collected across diverse sensor placements such as finger pulse oximeters, wrist-worn wearables, in-ear sensors, and multi-site configurations. This diversity allows evaluation of robustness and generalization under realistic deployment scenarios. We will delve into key parts in the dataset below.

3.2 Source Dataset Selection

First, PulseLM aims to integrate PPG recordings from publicly available datasets selected to maximize diversity in recording environment, sensor configuration, and supported physiological tasks. To provide an overall view of this diversity, a consolidated summary of dataset-level statistics, including QA scale, sensor placement, and environment, is provided in Table 1.

Clinical datasets form the backbone of PulseLM and provide high-quality physiological data. These datasets include hospital and perioperative monitoring data, intensive care recordings, and sleep laboratory studies, often with co-recorded reference signals such as ECG, arterial blood pressure, respiration, or polysomnography. As a result, they support QA tasks related to heart rate (HR), heart rate variability (HRV), blood pressure categorization (BP), atrial fibrillation detection (AF), multi-class arrhythmia classification (Arrhythmia), oxygen saturation (SpO2), respiratory rate (RR), sleep-disordered breathing severity (SDB), and signal quality index (SQI).

Laboratory datasets complement these clinical sources by capturing PPG signals under structured experimental protocols, such as controlled physical activities or emotional induction paradigms. These datasets contribute annotations for contextual and affective inference, including activity recognition and stress assessment (Stress), while maintaining moderate control over acquisition conditions and signal quality.

In-the-wild wearable datasets, in contrast, consist of long-duration PPG recordings collected during unconstrained daily activities using wearable devices. These recordings exhibit substantial variability in motion artifacts, environmental conditions, and sensor placement, and are therefore essential for evaluating model robustness and cross-domain generalization.

Across all sources, PulseLM covers finger-, wrist-, ear-, and multi-location PPG acquisition, as well as a wide range of device characteristics. Taken together, this diversity is a deliberate design choice to discourage overfitting to dataset-specific artifacts and to promote learning of transferable physiological representations.

3.3 Signal Standardization

To enable consistent multimodal learning across heterogeneous data sources, all PPG recordings are standardized through a unified preprocessing pipeline comprising four stages:

Resampling. Raw PPG signals are acquired at varying sampling rates depending on the source device and dataset protocol. To eliminate this variability, all recordings are resampled to 125 Hz.

Filtering. Next, a fourth-order Butterworth low-pass filter with a cutoff frequency of 8 Hz is applied to suppress high-frequency noise from motion artifacts, powerline interference, and sensor electronics. Subsequently, the DC component is removed by subtracting the per-segment mean, eliminating baseline drift caused by amplitude offsets. This two-stage filtering preserves the hemodynamically relevant AC component of the PPG waveform.

Table 1: Consolidated statistics of the PulseLM dataset constructed from fifteen public PPG sources.

No.	Dataset	# PPG Segments	# QA Cats	# QA Pairs	QA Categories	Sensor Positions	Environment
(1)	VitalDB [18]	163,959	5	819,795	HR, BP, HRV	finger	clinical
(2)	UCI [16]	111,751	3	335,253	HR, BP, SQI	finger	clinical
(3)	BCG [19]	671	3	2,013	HR, BP, SQI	finger	clinical
(4)	PPG-BP [20]	369	2	738	HR, BP	finger	clinical
(5)	SDB [10]	258,897	1	258,897	SDB	finger	clinical
(6)	Sensors [6]	2,061	3	6,183	HR, BP, SQI	finger	clinical
(7)	UQVitalSigns [21]	37,018	5	185,090	HR, BP, SpO_2, RR, SQI	finger	clinical
(8)	PPGArrhythmia [23]	46,827	1	46,827	Arrhythmia	finger	clinical
(9)	MIMIC PERform [8]	4,196	1	4,196	AF (Binary)	finger	clinical
(10)	EarSet [26]	1,776	1	1,776	HR	ear	lab
(11)	UTSA-PPG [42]	16,925	4	67,700	HR, HRV	wrist, finger	lab
(12)	WESAD [38]	2,998	1	2,998	Stress	wrist	lab
(13)	DALIA [34]	39,216	1	78,432	HR	wrist	in-the-wild
(14)	WildPPG [25]	240,000	4	960,000	HR, HRV	wrist, mix	in-the-wild
(15)	AFPPGECG [7]	384,001	1	384,001	AF (Binary)	wrist	mix
Total	**15 source datasets**	**1,310,665**	**12**	**3,153,899**	mix	mix	mix

Segmentation. After that, the filtered signals are partitioned into windows of 10 seconds, corresponding to 1,250 samples per segment. This window duration is commonly selected to capture approximately 8-15 cardiac cycles under normal conditions, providing sufficient temporal context for downstream tasks [32].

Normalization. Finally, each segment is independently normalized to the [0, 1] range using min-max scaling: $\tilde{x}_t = (x_t - x_{\min})/(x_{\max} - x_{\min})$. This per-segment normalization eliminates large absolute amplitude differences arising from sensor placement, skin tone, or contact pressure, while preserving the relative waveform morphology and temporal structure that encode physiological information.

3.4 Ground Truth Harmonization

In our study, PulseLM derives supervision from original dataset annotations or from reference signals co-recorded with PPG, such as ECG, arterial blood pressure. Depending on the source, available ground truth includes continuous-valued physiological measurements (e.g., heart rate, blood pressure, and HRV metrics), categorical clinical labels (e.g., atrial fibrillation or arrhythmia type), contextual states (e.g., activity or stress), and signal quality indicators. For datasets that provide synchronized ECG recordings, heart rate and heart rate variability labels are computed using established ECG-based signal processing toolkits [24], treating ECG as the physiological gold standard for rhythm and variability estimation. We show the label distributions across tasks in Figure 3.

Next, to enable unified language-based supervision across heterogeneous datasets, all ground truth is harmonized into a shared categorical label space using predefined mapping rules. Continuous-valued measurements are discretized into clinically or physiologically meaningful categories (e.g., bradycardia/normal/tachycardia for heart rate, standard risk-hypertension stages for blood pressure), including ECG-derived HR and HRV metrics where applicable. Dataset-provided categorical labels are preserved when compatible and otherwise normalized to a consistent label set. All mapping rules and thresholds are fixed *a priori* and applied deterministically; the complete specifications are provided in Appendix A.2. Through this process, PulseLM supports twelve closed-ended QA tasks spanning cardiovascular state, rhythm analysis, variability assessment, signal quality, behavior and affect, respiratory function, and sleep-related risk. All QA supervision is grounded in original annotations or reference signals, and language models infer no labels during dataset construction.

3.5 Question Answering Formulation

PulseLM represents physiological supervision using a closed-ended question answering (QA) formulation, in which each PPG segment is paired with natural-language questions and answers drawn from fixed, clinically meaningful label sets. For each QA category, a predefined answer vocabulary is specified based on the harmonized physiological labels described in Section 3.4. This formulation constrains model outputs to a finite and interpretable set of responses, enabling consistent supervision and objective evaluation across heterogeneous datasets.

To introduce linguistic variability while maintaining semantic consistency, questions are generated offline for each QA category using controlled templates and paraphrasing. A large language model is employed solely to produce semantically equivalent rephrasings of these templates. The generation process is restricted to question text; all answers are deterministically assigned by mapping existing ground-truth annotations to the corresponding categorical labels. As a result, the QA construction procedure does not introduce new physiological information or alter the underlying supervision.

Depending on the availability of annotations in the source dataset, a single PPG segment may be associated with multiple QA instances spanning different physiological domains. The QA statistics reported in PulseLM reflect the realized QA pairs instantiated for each segment, rather than hypothetical combinations of questions and labels. This design yields dense, multi-task supervision while preserving a direct and verifiable correspondence between each QA instance and its underlying physiological ground truth.

At this end, PulseLM supports twelve QA tasks covering cardiovascular state, rhythm analysis, variability assessment, signal quality, behavior and affect, respiratory function, and sleep-related risk. The final release comprises 1,310,665 standardized PPG segments paired with 3,153,899 QA instances. Dataset-level QA coverage, sensor placement, and recording environment are summarized in Table 1. All benchmarks defined in this work operate at the segment and QA level.

4 PulseLM Benchmarking

4.1 Objectives and Scope

This section establishes standardized benchmarks for *multimodal physiological question answering* from PPG signals using the PulseLM dataset. Unlike prior PPG studies that focus on narrowly defined regression or classification tasks, PulseLM is designed to evaluate whether models can jointly reason over continuous wearable signals and natural language queries within a unified framework.

The benchmarking objectives are twofold. First, we aim to provide reproducible baselines for training and evaluating *large multimodal PPG-language models* that integrate raw PPG signals with natural-language questions. Second, we investigate whether such models can learn transferable physiological representations that generalize across datasets with different sensor placements, acquisition conditions, and annotation types. To this end, all benchmarks are defined at the QA level and operate over the harmonized label space described in Section 3.

4.2 Training and Evaluation Protocol

For multimodal fine-tuning, models are trained to condition their responses jointly on the PPG signal representation and the input question. During evaluation, the model is prompted with a PPG segment and a single question and is required to select one answer from the predefined candidate set. Performance is measured primarily using a strict exact-match classification accuracy, averaged across question paraphrases to reduce sensitivity to surface-level linguistic variation.

To probe robustness and generalization, we define both **in-dataset** and **cross-dataset** evaluation settings. In the in-dataset setting, training and evaluation are conducted on splits drawn from the same source datasets. In the cross-dataset setting, models are fine-tuned on a dataset and evaluated on unseen datasets with different sensor placements, recording environments, and population characteristics. This setting explicitly tests whether models learn transferable physiological representations rather than dataset-specific heuristics. All benchmarks use fixed train-validation-test splits and consistent QA templates to ensure reproducibility. Detailed split definitions and evaluation scripts are released with the dataset.

4.3 Baseline Models

To establish reference performance levels on PulseLM, we benchmark a family of multimodal physiological language models that explicitly integrate PPG signals with LLMs, as shown in Figure 2. Here, each baseline follows a unified *PPG-Text LLM* architecture that combines a dedicated PPG encoder with an instruction-tuned language backbone, enabling conditional reasoning over continuous biosignals and natural-language queries.

PPG Encoder. First, raw PPG waveforms are processed by a pretrained PPG encoder that operates directly on the time-series signal. Given a 10-second PPG segment, the encoder produces a sequence of latent PPG embeddings that capture waveform morphology and temporal dynamics. We evaluate multiple PPG encoders with different inductive biases, while keeping the downstream language model fixed to isolate the effect of physiological representation learning.

Mapping and Token Alignment. Since the dimensionality of PPG encoder outputs typically differs from that of the LLM token embeddings, we introduce a lightweight *mapping layer* to project PPG embeddings into the LLM embedding space. This mapping layer consists of a linear projection (optionally followed by normalization) that transforms each PPG embedding into a token-compatible representation. The resulting projected embeddings are treated as a sequence of *PPG prefix tokens*.

Multimodal Fusion and Decoding. The projected PPG prefix tokens are concatenated with the embedded text tokens derived from the input question. Specifically, the question is first tokenized and embedded using the LLM's standard tokenizer and token embedding layer. The PPG prefix tokens are then prepended to the text token sequence, forming a unified multimodal input that is fed into the LLM decoder. Here, we conduct experiments with a diverse set of instruction-tuned and domain-adapted LLM backbones spanning multiple model scales. This design allows the language model to attend jointly over physiological and textual information using its native self-attention mechanism, without modifying the LLM architecture. Eventually, given the concatenated PPG and text tokens, the LLM decoder generates an answer token sequence conditioned on both modalities.

4.4 Implementation Details

All baseline models are trained under a unified optimization and evaluation protocol to ensure fair and reproducible comparison. We evaluate multiple instruction-tuned large language models that span a broad range of model capacities, including Qwen3-4B-Instruct-2507 [40], LLaMA-3.1-8B-Instruct, LLaMA-3.2-3B-Instruct, and LLaMA-3.2-1B-Instruct[1]. The PPG encoder is initialized from a pretrained Papagei-S model[2] and kept frozen throughout training to preserve its pretrained physiological representations, while the projection layer is learned jointly with the language model. Specifically, we fine-tune the language backbones using low-rank adaptation (LoRA) [14], with rank $r = 64$ and scaling factor $\alpha = 128$, enabling stable adaptation across different model sizes while keeping the number of trainable parameters are manageable within our available resources. Finally, Fine-tuning is performed for one epoch using a learning rate of 1×10^{-4}, a warm-up ratio of 0.03.

4.5 Main Results

Benchmark results are summarized in Tables 2-4, reporting exact-match (EM) accuracy under in-domain evaluation across task categories (Table 2) and across the 15 constituent datasets (Table 3), as

[1] https://huggingface.co/meta-llama/models
[2] https://github.com/Nokia-Bell-Labs/papagei-foundation-model

(pretty nice scaling)

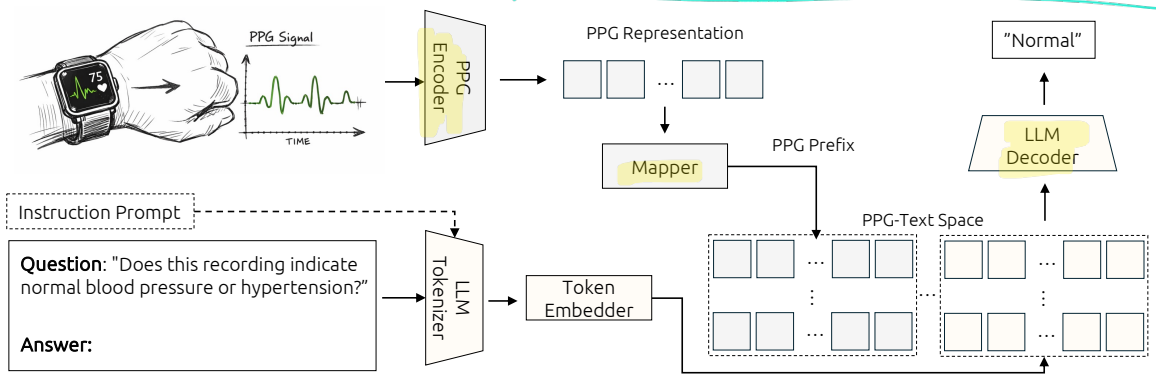

Figure 2: Illustration of benchmarking PPG language modeling.

Table 2: In-domain benchmark results on PulseLM using exact-match (EM) accuracy across different tasks.

Model	HR	BP	HRV	RR	AF	Arrhythmia	SpO_2	SDB	Stress	SQI	Avg.
LLaMA3.2-1B	0.03	0.03	0.01	0.01	0.74	0.38	0.01	0.51	0.12	0.07	0.19
LLaMA3.2-3B	0.16	0.14	0.16	0.37	0.88	0.21	0.37	0.12	0.01	0.19	0.26
LLaMA3-8B	0.55	0.52	0.55	0.78	0.87	0.88	0.78	0.58	0.36	0.53	0.64
Qwen3-4B	0.53	0.48	0.51	0.65	0.54	0.90	0.65	0.79	0.74	0.53	0.63

Table 3: In-domain benchmark results on PulseLM using exact-match (EM) accuracy across different datasets. Columns correspond to individual datasets indexed by (1)-(15) following the order in A.1.

Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	Avg.
LLaMA3.2-1B	0.01	0.10	0.13	0.3	0.51	0.32	0.01	0.38	0.20	0.50	0.01	0.12	0.20	0.01	0.78	0.23
LLaMA3.2-3B	0.12	0.10	0.25	0.01	0.12	0.14	0.37	0.21	0.56	0.10	0.28	0.01	0.15	0.25	0.90	0.24
LLaMA3-8B	0.51	0.41	0.43	0.45	0.58	0.51	0.78	0.88	0.61	0.78	0.61	0.36	0.52	0.68	0.89	0.60
Qwen3-4B	0.45	0.48	0.41	0.75	0.79	0.55	0.65	0.90	0.78	0.95	0.64	0.74	0.67	0.69	0.53	0.66

Table 4: Cross-dataset generalization: models trained on the VitalDB dataset and evaluated on other unseen datasets for heart rate (HR) and blood pressure (BP) classification. Results reported as exact-match (EM) accuracy (%).

	BCG		PPGBP		Sensors		UCI		
Model	HR	BP	HR	BP	HR	BP	HR	BP	Avg.
LLaMA3-8B	0.70	0.36	0.81	0.23	0.82	0.49	0.84	0.48	0.59
Qwen3-4B	0.84	0.25	0.90	0.38	0.97	0.23	0.91	0.31	0.59

well as cross-dataset generalization when training on VitalDB and testing on unseen datasets for HR/BP classification (Table 4).

In-domain performance across tasks: As shown in Table 2, model capacity has a substantial impact on performance. LLaMA3-8B and Qwen3-4B achieve the strongest overall results, with average accuracies of 0.64 and 0.63, respectively, significantly outperforming LLaMA3.2-3B (0.26) and LLaMA3.2-1B (0.19). Across task categories, rhythm-related tasks such as AF and Arrhythmia achieve the highest average accuracy, while tasks requiring finer-grained physiological estimation or signal assessment, such as HR, BP, HRV, Stress, and SQI remain more challenging. LLaMA3-8B and Qwen3-4B exhibit complementary strengths across tasks, with LLaMA3-8B performing better on several physiological estimation tasks and Qwen3-4B achieving higher accuracy on tasks such as SDB and Stress.

In-domain performance across datasets: Table 3 reports in-domain results across the 15 constituent datasets. Consistent with task-level results, Qwen3-4B achieves the highest overall average accuracy (0.66), followed by LLaMA3-8B (0.60), while smaller models remain substantially less competitive. Both strong models show noticeable performance variation across datasets, suggesting that differences in sensing conditions, populations, and annotation characteristics continue to pose challenges even under a unified QA formulation.

Cross-dataset generalization: Table 4 evaluates cross-dataset generalization by training models on VitalDB and testing on unseen datasets. LLaMA3-8B and Qwen3-4B achieve comparable average performance (0.59 EM), demonstrating similar overall transfer capability. Across all target datasets, HR generalizes more robustly than

(the 12 QA tasks)

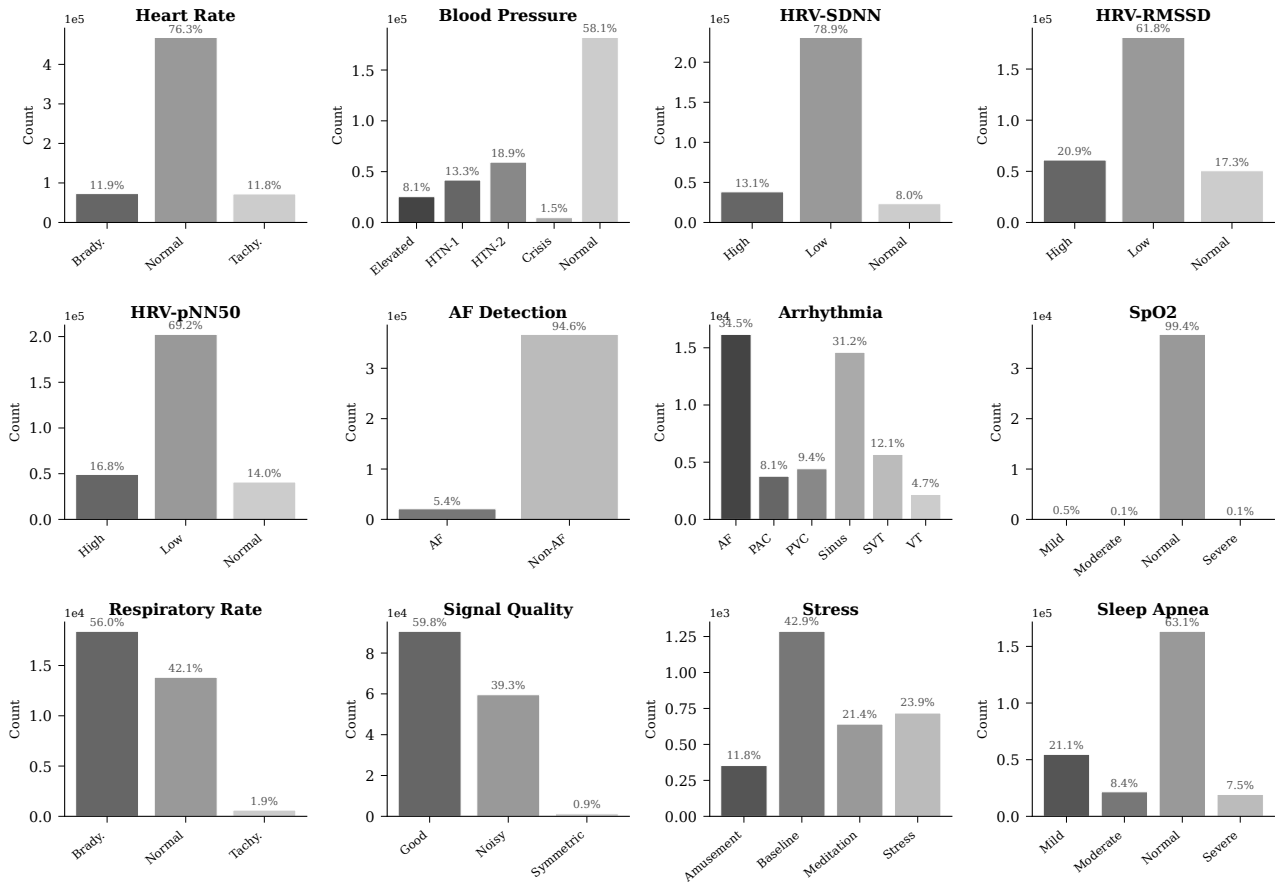

Figure 3: Demonstration of label distributions in PulseLM dataset.

BP, highlighting BP classification as a more domain-sensitive and challenging task. While Qwen3-4B shows consistently strong HR transfer, LLaMA3-8B performs comparatively better on several BP targets, underscoring differing generalization behaviors between the two backbones.

5 Discussion

Despite its scale and breadth, PulseLM has several points that merit discussion. First, although all answer labels are deterministically derived from original dataset annotations or reference signals (e.g., ECGs), a portion of the natural language content used during question construction is reasonably generated through automated or semi-automated procedures to mimic how different users may use their different prompts during inference. While we employ controlled paraphrasing and validation checks to ensure semantic consistency, this process may still introduce subtle linguistic bias or distributional artifacts. Second, although PulseLM spans a wide range of physiological attributes, sensor placements, and recording environments, it is intended for research and benchmarking purposes and does not replace directly clinically validated measurements or expert diagnosis.

At the same time, PulseLM opens several promising directions for future work. One natural extension is the incorporation of additional expert-verified annotations and higher-level clinical summaries to further enhance interpretability and linguistic fidelity. Beyond closed-ended QA, PulseLM also includes structured textual information derived from original dataset metadata—such as physiological labels, demographic attributes, activity context, sensor placement, and recording environment—which provides a foundation for exploring open-ended physiological report generation. In this setting, models could be trained to produce free-form textual summaries grounded in PPG signals and contextual cues, enabling richer and more flexible interaction than discrete QA alone.

Another important direction is multimodal representation learning, where textual supervision from QA pairs or generated reports is leveraged to guide self-supervised or weakly supervised learning of transferable PPG representations. Such approaches may improve robustness across tasks, datasets, and sensing conditions. Finally, expanding cross-domain generalization benchmarks and developing confidence-aware generation mechanisms remain critical steps toward deploying physiological language models in safety-sensitive health monitoring applications.

6 Conclusion

This paper presents PulseLM, a large-scale dataset and benchmark designed to support multimodal physiological reasoning from PPG signals using language-based supervision. By consolidating heterogeneous PPG recordings from clinical, laboratory, and in-the-wild settings into a unified closed-ended question-answering formulation, PulseLM enables consistent modeling and evaluation across diverse physiological tasks that are otherwise studied in isolation. The accompanying baseline results highlight both the promise and current limitations of multimodal physiological LLMs under realistic domain shifts. We expect PulseLM to serve as a reference point for future work on scalable physiological foundation models, cross-dataset generalization, and interpretable reasoning over wearable biosignals.

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A Appendix

A.1 Source Dataset Details

In our study, we utilize various of public PPG datasets as the sources to construct the QA dataset. In this section, we will introduce the datasets we use.

UCI [16]: The UCI dataset contains high-resolution PPG signals extracted from thousands of clinical records within the MIMIC-II database. These signals are synchronized with invasive arterial blood pressure waveforms at a sample rate of 125 Hz, primarily using finger-based transmittance sensors. This collection is essential for training models that map PPG morphology to systolic and diastolic pressure values.

PPG-DaLiA [35]: PPG-DaLiA provides physiological data from 15 participants performing a wide range of daily life activities, recorded at a sample rate of 64 Hz. The PPG data is collected from a Empatica E4 wristband and is accompanied by accelerometer data to facilitate the removal of motion artifacts. The dataset contains 64,697 samples from two hours of recordings per subject.

WESAD [38]: The WESAD dataset features PPG recordings from both the wrist and the chest, collected at a frequency of 64 Hz during controlled emotional stress experiments. It includes signals like PPG, ECG, and accelerometer, capturing neutral, stress, and amusement states. It serves as a key resource for developing mental health monitoring applications using wearable sensors. The dataset provides 43,385 samples from 100 minutes of recordings per participant.

BCG [19]: The BCG dataset features PPG signals recorded at 140 Hz from 32 subjects to establish a reliable ground truth for cardiac timing. These signals are captured using finger-mounted sensors to correlate mechanical body vibrations with the optical pulse wave during rest. The collection encompasses multiple nights of continuous data recorded in the participants' regular sleep environments using piezoelectric film sensors. By synchronizing these optical and mechanical measurements, the dataset facilitates the validation of non-contact physiological monitoring systems.

PPG-BP [20]: The PPG-BP dataset consists of PPG recordings from 219 participants collected at a high sampling rate of 1000 Hz. During the data acquisition procedure, three distinct PPG signal segments were recorded for each subject. Each individual segment contains 2100 samples, providing high-resolution pulse wave morphology for analysis. This structure allows for a robust assessment of arterial features across a diverse group of subjects.

SDB [10]: The SDB dataset comprises PPG recordings from 146 children undergoing overnight sleep studies to identify patterns of disordered breathing. These physiological signals were captured over six hours at a sampling rate of 62.5 Hz using a specialized Phone Oximeter device. Participants are classified into positive or negative groups according to their Apnea-Hypopnea Index scores, which serve to quantify the severity of obstructive sleep apnea. This specific focus on nocturnal pulse waves makes the dataset a valuable resource for studying respiratory-induced variations in the PPG signal.

Sensors [6]: Sensors is a curated subset of MIMIC-III that contains ICU records from 1,195 patients. It provides paired PPG and arterial blood pressure (ABP) waveforms collected via the Philips CareVue Clinical Information System and the iMDsoft MetaVision ICU platform. A key characteristic of this dataset is its discrete segmentation strategy: for each record, the authors retain only two 15-second segments separated by 5 minutes. As a result, Sensors offers a medium-to-large collection with substantial inter-subject variability, a reasonable number of segments per subject, and a standardized, sparse sampling protocol that is well-suited for robust PPG-to-BP modeling and benchmarking.

UQVitalSigns [21]: The UQVitalSigns dataset is a high-resolution, multi-parameter anesthesia monitoring repository collected from 32 surgical cases at the Royal Adelaide Hospital, covering case durations from 13 minutes to 5 hours (median 105 minutes). It provides time-synchronized vital signs and waveforms captured from Philips IntelliVue MP70/MP30 patient monitors and Datex-Ohmeda Aestiva/5 anesthesia machines, including signals commonly used in perioperative monitoring such as the pulse oximeter pleth waveform (PPG), ECG, capnography, noninvasive blood pressure, and airway flow/pressure (with additional channels available in some cases). The authors align and interpolate measurements to 10 ms resolution and release the data in accessible formats (e.g., full waveform/numerical exports and visualization files).

UTSA-PPG [42]: The UTSA-PPG dataset was developed to support comprehensive heart rate and heart rate variability studies by addressing the lack of unified, all-in-one physiological datasets. It features data from 12 subjects across 36 individual sessions, with each session providing 30 minutes of continuous recording sampled at 100 Hz. The PPG signals are captured simultaneously from both the wrist and the fingers, allowing for high-resolution analysis of pulse wave morphology across different body sites.

WildPPG [25]: WildPPG offers unique PPG data collected in unconstrained, real-world environments over several hours per subject. The dataset features PPG signals recorded at 128Hz during various outdoor activities from sixteen participants. These signals were captured using four wearable sensors placed at different body sites, including the forehead and wrist, to study signal reliability outside of a laboratory setting.

EarSet [26]: The EarSet dataset investigates the ear canal as a stable site for PPG sensing, providing data from in-ear wearables sampled at 100 Hz. Earset collectes a multi-modal dataset from 30 participants while performing 16 natural motions, including both head/face and full body movements.

VitalDB [18]: VitalDB is a massive intraoperative dataset containing high-fidelity PPG waveforms recorded at 500 Hz from thousands of surgical cases. The PPG signals are collected from the patients' fingers. It provides a rich source of data for analyzing how the PPG signal changes under anesthesia and during acute hemodynamic shifts. The dataset consists of a total of 557,622 (average 87, range 16-136) data tracks from 6,388 cases.

PPGArrhythmia [23]: The PPGArrhythmia dataset is a clinical PPG benchmark for multiclass arrhythmia detection, providing 10-second PPG segments sampled at 100 Hz (i.e., 1,000 samples per segment) with labels for six rhythm classes: sinus rhythm, PVC, PAC, VT, SVT, and atrial fibrillation. It includes data from 228 patients with 118,217 labeled 10-second PPG segments.

MIMIC PERform [8]: The MIMIC PERform dataset (MIMIC PERform AF) is a clinically collected PPG resource extracted from the MIMIC-III Waveform Database (Matched Subset), consisting of 20-minute recordings from 35 critically ill adult subjects acquired

during routine ICU care. It provides synchronized PPG, ECG, and respiration signals recorded from a bedside monitor at 125 Hz, and includes subject-level labels indicating whether each recording was captured during atrial fibrillation (19 subjects) or normal sinus rhythm / non-AF (16 subjects), making it a practical benchmark for AF-related analysis and robust PPG algorithm evaluation in real clinical conditions.

AFPPGECG [7]: The AFPPGECG dataset provides long-term, simultaneously acquired wrist PPG and reference ECG recordings for atrial fibrillation (AF) research, with manually verified AF episode annotations. It includes data from 8 subjects with suspected AF monitored for 5–8 days (1,306 hours total). The wrist PPG is recorded at 100 Hz (green LED), while the reference ECG is recorded at 500 Hz; the dataset also includes triaxial accelerometry (50 Hz on the wrist device and 25 Hz on the ECG device), enabling studies on motion effects and robust AF detection in real-world long-duration monitoring.

A.2 Label Mapping Harmonization

As the included PPG sources provide heterogeneous annotations (e.g., continuous values such as heart rates), we harmonize all physiological measurements into a unified set of clinically interpretable categorical labels. This mapping (i) standardizes supervision across datasets with different acquisition settings and label conventions, and (ii) enables closed-ended physiological QA with a fixed answer space per category. When a measurement is unavailable for a sample (e.g., no BP or no AHI), we omit the corresponding QA category for that sample.

Heart rate (HR). We discretize HR (bpm) into three classes: `bradycardia` (< 60), `normal` (60–100), and `tachycardia` (> 100).

Blood pressure (BP). We map SBP/DBP (mmHg) into five risk categories, using a severity-prioritized decision order: `hypertensive crisis` (SBP> 180 **or** DBP> 120), `hypertension stage 2` (SBP≥ 140 **or** DBP≥ 90), `hypertension stage 1` (SBP≥ 130 **or** DBP≥ 80), `elevated` (SBP 120–129 **and** DBP< 80), and `normal` (SBP< 120 **and** DBP< 80).

Signal quality (SQI via skewness). We use a skewness-based heuristic to discretize waveform quality. Segments with skewness in $[0.5,\ 2.0]$ are labeled `good_quality`; otherwise they are labeled `noisy_or_distorted`. Additionally, nearly symmetric waveforms (skewness approximately 0, implemented with a small tolerance) are flagged as `symmetric_unusual`, as such segments may indicate atypical morphology or preprocessing artifacts.

Sleep-disordered breathing (SDB via AHI). We discretize AHI (events/hr) into `normal_ahi<5` (< 5), `mild_5<=ahi<15` (5–14), `moderate_15<=ahi<30` (15–29), and `severe_ahi>=30` (≥ 30).

Heart rate variability (HRV). We discretize SDNN (ms) into `low` (< 50), `normal` (50–100), `high` (> 100); RMSSD (ms) into `low` (< 20), `normal` (20–50), `high` (> 50); and pNN50 (%) into `low` (< 3), `normal` (3–25), `high` (> 25).

SpO_2. We discretize SpO_2 (%) into `normal` (≥ 95), `mild_hypoxemia` (90–94), `moderate_hypoxemia` (85–89), and `severe_hypoxemia` (< 85).

Respiratory rate (RR). We map RR (breaths/min) into `bradypnea` (< 12), `normal` (12–20), and `tachypnea` (> 20).